

Cybersecurity

Initial Access and Internal Recon

Lesson 15.2



Learning Objectives

By the end of today's class, you will be able to:

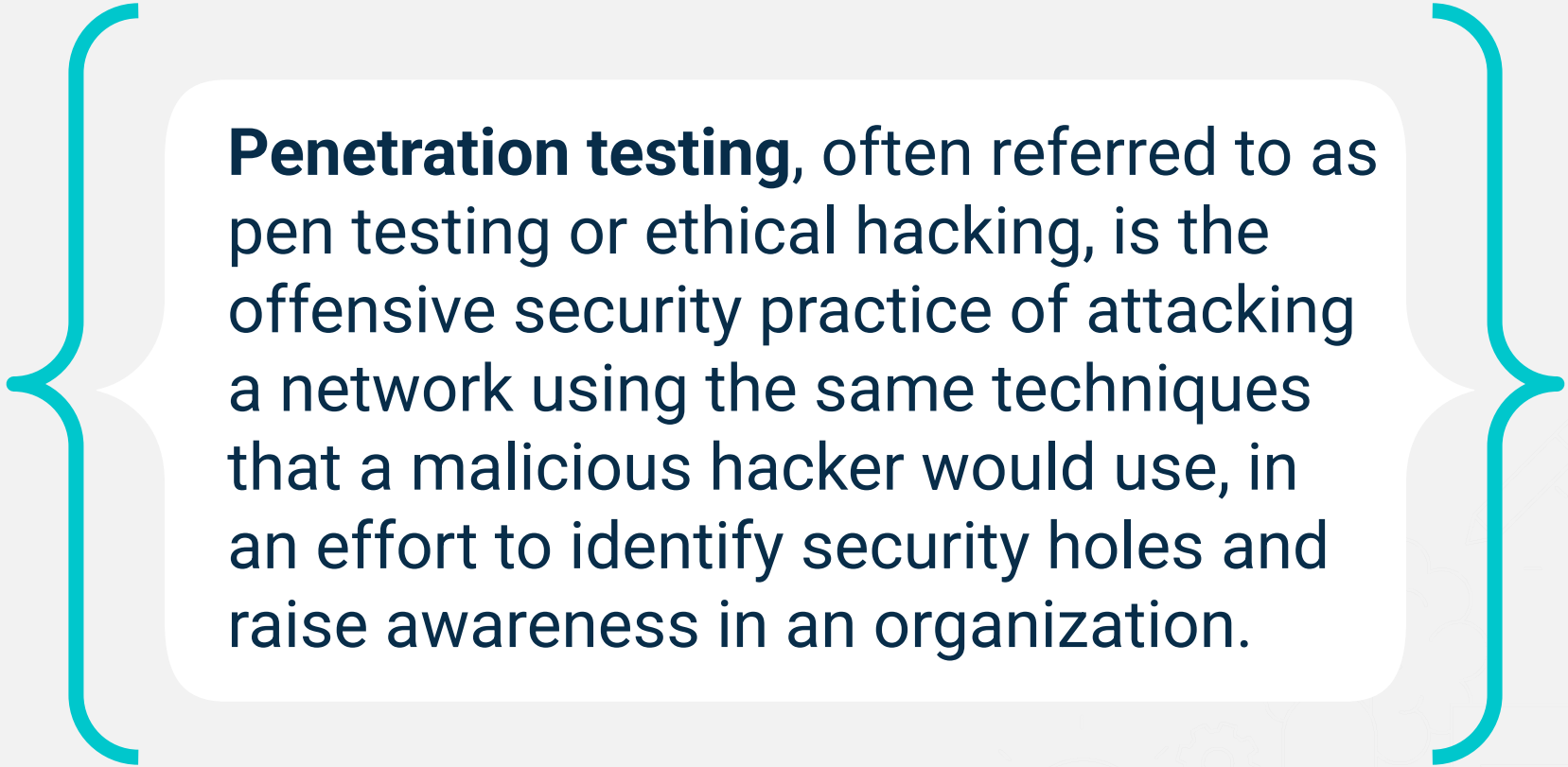
- 1 Understand how initial access fits into the MITRE matrix.
- 2 Recognize phishing emails and understand why attackers so commonly use them in order to obtain initial access.
- 3 Perform advanced Nmap scans with NSE scripts.
- 4 Exploit a machine with a Python script.



Welcome



Let's recap



Penetration testing, often referred to as pen testing or ethical hacking, is the offensive security practice of attacking a network using the same techniques that a malicious hacker would use, in an effort to identify security holes and raise awareness in an organization.



Day 1 Recap

The five phases of a pen testing engagement include:

01A

Planning

Define the purpose and scope of the test, and sign all legal contracts.

01B

Reconnaissance

Obtain publicly available information about your target.

02

Scanning

Use tools to run a scan against your target to gather information, such as open ports, and run services to determine potential vulnerabilities.

03

Exploitation

Attack the vulnerabilities discovered in the previous steps to gain access to the target.

04

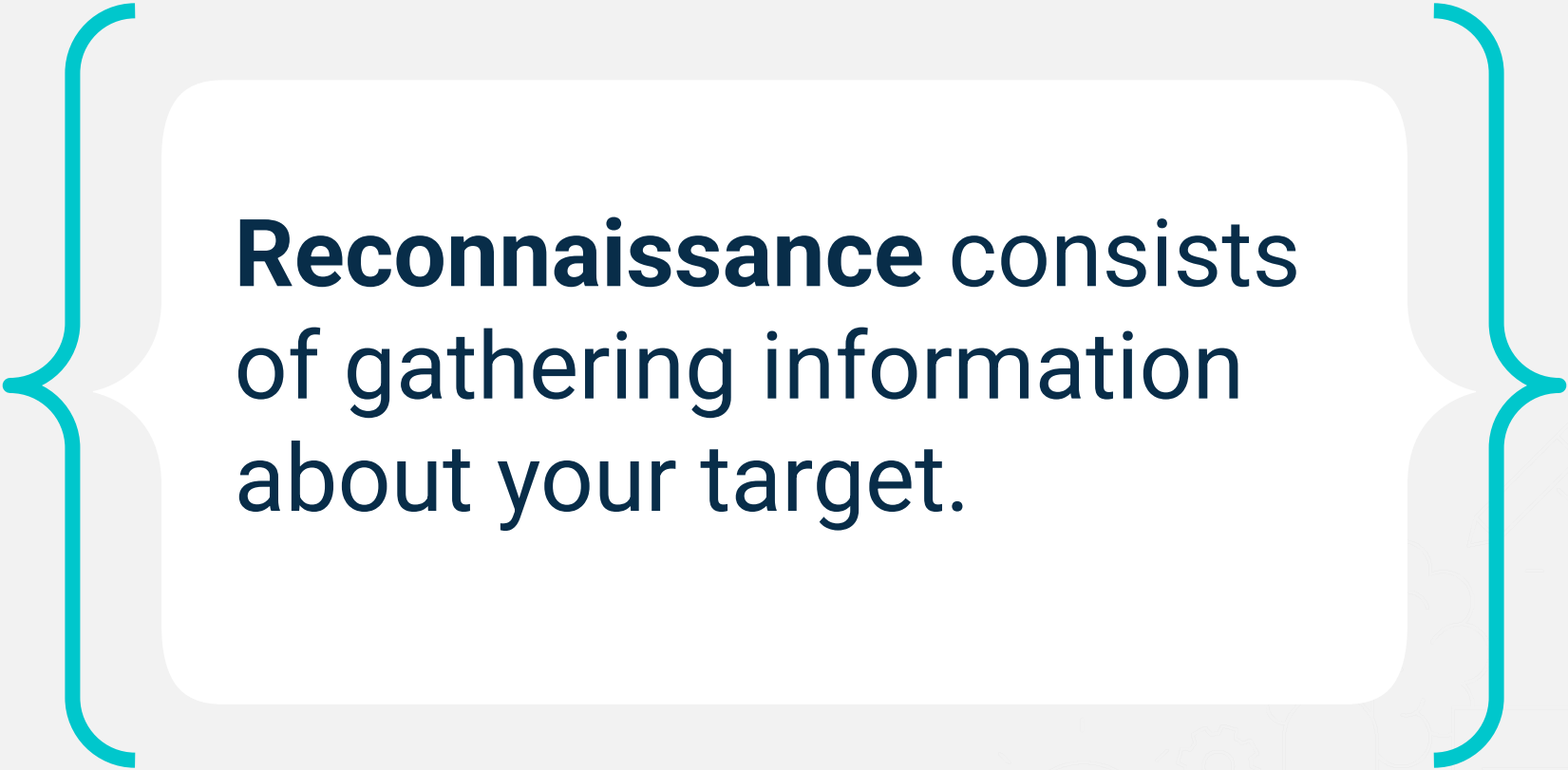
Post exploitation

Gather valuable information from the compromised systems.

05

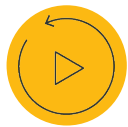
Reporting

Report on the previous five steps to provide a summary of actions taken, findings, and recommended mitigations.



Reconnaissance consists
of gathering information
about your target.





Day 1 Recap

Reconnaissance is divided into two types:

01

Passive Recon

Often refers to **open source intelligence (OSINT)**, which leverages information about the target that is publicly available on the internet.

This includes all domains and hosts belonging to a target that are publicly viewable.

02

Active Recon

Also refers to gathering information about the target, but it involves directly interacting with the target.



Day 1 Recap

The **MITRE ATT&CK® Matrix** provides a visual representation of all the **techniques**, **tactics**, and **procedures (TTPs)** that may be performed throughout an assessment.

We will use this matrix as a reference for the TTPs that we will conduct throughout the next two weeks.

MITRE | ATT&CK®

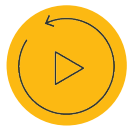
Home > Matrices > Enterprise

Enterprise Matrix

Below are the tactics and techniques representing the MITRE ATT&CK® Matrix for Enterprise. The Matrix contains information for the following platforms: Windows, macOS, Linux, PRE, Azure AD, Office 365, Google Workspace, SaaS, IaaS, Network, Containers.

layout: side ▾ show sub-techniques hide sub-techniques help

Tactic	Techniques
Reconnaissance	10 techniques
Resource Development	7 techniques
Initial Access	9 techniques
Execution	12 techniques
Persistence	19 techniques
Privilege Escalation	13 techniques
Defense Evasion	39 techniques
Credential Access	15 techniques
Discovery	27 techniques
Lateral Movement	9 techniques
Collection	17 techniques



Day 1 Recap



A reconnaissance tactic in which we manipulate **Google** searches to narrow down our queries to acquire actionable intel.



Day 1 Recap

Certificate transparency is another reconnaissance tactic whereby a penetration tester gathers information from certificate issuers, which publish logs of the SSL/TLS certificates that they issue to organizations.

The screenshot shows the crt.sh Identity Search interface. The search criteria are set to 'Identity' with a match of 'ILIKE' and a search term of 'example.com'. The results table lists certificates with columns for crt.sh ID, Logged At, Not Before, Not After, Matching Identities, and Issuer Name.

Certificates	crt.sh ID	Logged At	Not Before	Not After	Matching Identities	Issuer Name
	987119772	2018-11-29	2018-11-28	2020-12-02	example.com	C=US, O=DigiCert Inc, CN=DigiCert SHA2 Secure Server CA
	984858191	2018-11-28	2018-11-28	2020-12-02	www.example.com	C=US, O=DigiCert Inc, CN=DigiCert SHA2 Secure Server CA
	34083306	2016-09-23	2010-09-02	2011-10-01	subjectname@example.com	emailAddress=pk_i_admin@sungard.com, O=SunGard Availability Services, CN=SAS Public CA v1
	34001389	2016-09-23	2010-09-02	2011-10-01	subjectname@example.com	emailAddress=pk_i_admin@sungard.com, O=SunGard Availability Services, CN=SAS Public CA v1
	24564717	2016-07-14	2016-07-14	2017-07-14	example.com	C=US, O="hawte, Inc.", CN=hawte SSL CA - G2
	24560643	2016-07-14	2016-07-14	2018-07-14	m.testexample.com	C=US, O="hawte, Inc.", CN=hawte SSL CA - G2
	24560621	2016-07-14	2016-07-14	2017-07-14	www.example.com	C=US, O="hawte, Inc.", CN=hawte SSL CA - G2
	24558997	2016-07-14	2016-07-14	2018-07-14	example.com	C=US, O=Symantec Corporation, OU=Symantec Trust Network, CN=Symantec Class 3 Secure Server CA - G4
	10557607	2015-11-05	2015-11-03	2018-11-28	dev.example.com	C=US, O=DigiCert Inc, OU=www.digicert.com, CN=DigiCert SHA2 High Assurance Server CA
	5857507	2014-12-11	2014-11-06	2015-11-13	products.example.com	C=US, O=DigiCert Inc, OU=www.digicert.com, CN=DigiCert SHA2 High Assurance Server CA
					support.example.com	
					www.example.com	

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Day 1 Recap

Shodan.io is a reconnaissance website that conducts port scanning across the entire internet and catalogs the results for quick searching.

The screenshot shows the Shodan.io website interface. The browser address bar displays 'shodan.io/host/93.184.216.34'. The website header includes navigation links like 'Shodan', 'Developers', 'Monitor', and 'View All...'. A search bar with the Shodan logo is prominent. Below the header, a satellite map of the Norwell area is shown. The main content area displays search results for the IP address 93.184.216.34. On the left, a table lists metadata: City (Norwell), Country (United States), Organization (Verizon Business), ISP (Verizon Business), Last Update (2020-04-24T17:31:40.422234), and ASN (AS15133). On the right, the 'Ports' section shows two open ports: 80 and 443. Below this, the 'Services' section lists the protocols (http, tcp) and a detailed HTTP response header for port 80, including status '200 OK', age, cache-control, content-type, date, etag, expires, last-modified, server, vary, x-cache, and content-length.

93.184.216.34	
City	Norwell
Country	United States
Organization	Verizon Business
ISP	Verizon Business
Last Update	2020-04-24T17:31:40.422234
ASN	AS15133

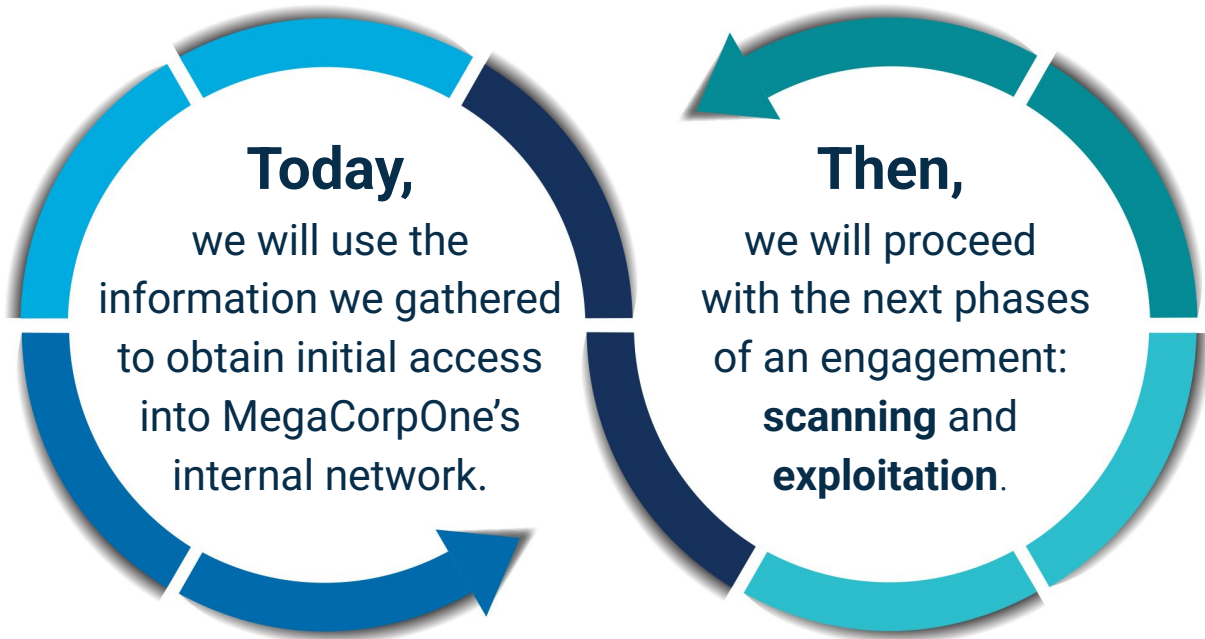
Ports	
80	443

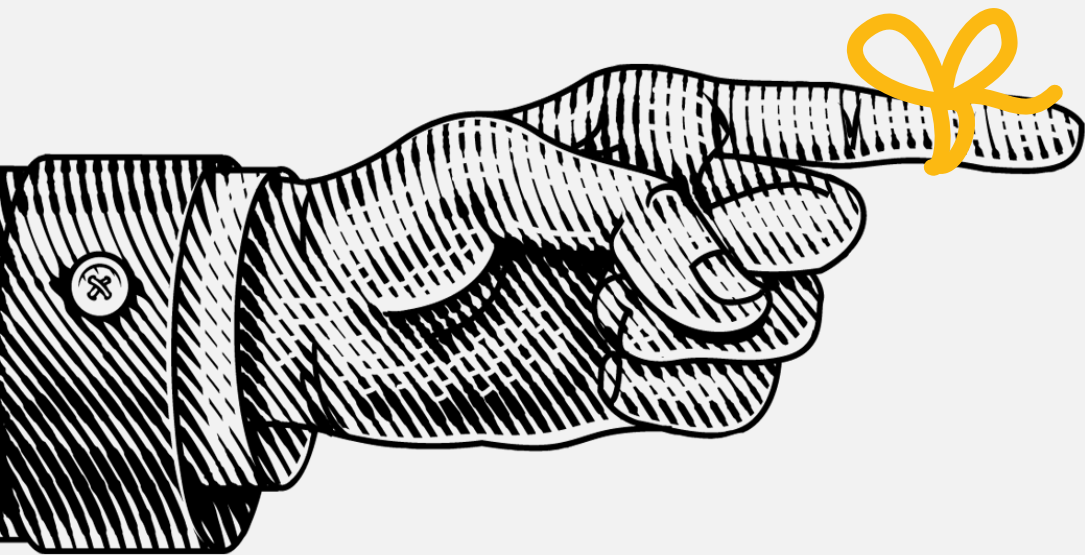
Services	
80	HTTP/1.1 200 OK
tcp	Age: 354667
http	Cache-Control: max-age=604800
	Content-Type: text/html; charset=UTF-8
	Date: Fri, 24 Apr 2020 17:31:40 GMT
	Etag: "3147526947+ident"
	Expires: Fri, 01 May 2020 17:31:40 GMT
	Last-Modified: Thu, 17 Oct 2019 07:18:26 GMT
	Server: ECS (bsa/EB11)
	Vary: Accept-Encoding
	X-Cache: HIT
	Content-Length: 1256



Day 1 Recap

We ended Day 1's lesson by using Shodan and Recon-ng in the **reconnaissance** phase to identify subdomains and other related domains belonging to MegaCorpOne.





Remember,

Every method we use this week will tie back to a specific tactic and technique found on the **MITRE ATT&CK® Matrix**.





Gaining Initial Access

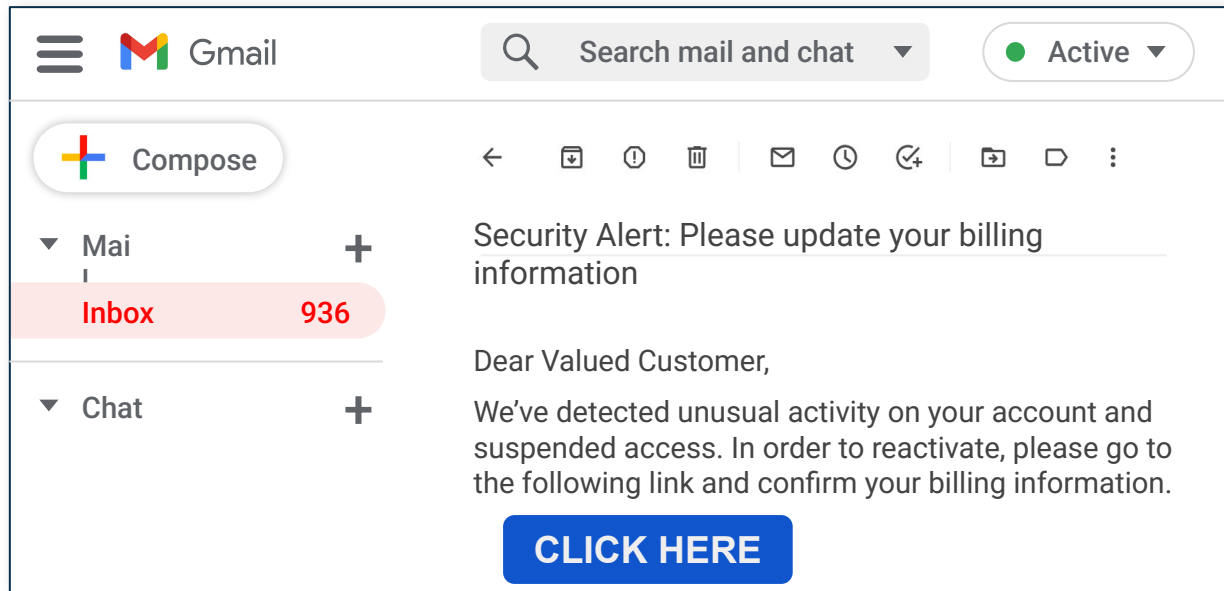
Gaining Initial Access

The information that we gathered in the previous class can be used to gain **initial access** into our target.

For example:

Through Google hacking, you may find an employee's name, job title, direct reports, and email address.

You could use that information to craft a deceptive email to try and obtain the employee's credentials for their private network.





MITRE ATT&CK®

Initial access consists of techniques that use various entry vectors to gain their initial foothold within a network.

Techniques used to gain a foothold include targeted spear phishing and exploiting weaknesses on public-facing web servers.

Footholds gained through initial access may allow for continued access, like valid accounts and use of external remote services, or may be limited-use due to changing passwords.

Gaining Initial Access

Initial access is
Tactic ID TA0001.

MITRE | ATT&CK®

Matrices

Tactics ▾

Techniques ▾

Mitigations ▾

Groups

Software

Resources ▾

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TACTICS

Enterprise ^

Reconnaissance

Resource

Development

Initial Access

Execution

Persistence

Privilege Escalation

Defense Evasion

Credential Access

Discovery

Lateral Movement

Collection

Command and Control

Exfiltration

Home > Tactics > Enterprise > Initial Access

Initial Access

The adversary is trying to get into your network.

Initial Access consists of techniques that use various entry vectors to gain their initial foothold within a network. Techniques used to gain a foothold include targeted spearphishing and exploiting weaknesses on public-facing web servers. Footholds gained through initial access may allow for continued access, like valid accounts and use of external remote services, or may be limited-use due to changing passwords.

ID: TA0001

Created: 17 October 2018

Last Modified: 19 July 2019

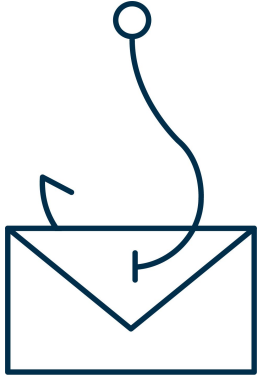
[Version](#) [Permalink](#)

Techniques

Techniques: 9

ID	Name	Description
T1189	Drive-by Compromise	Adversaries may gain access to a system through a user visiting a website over the normal course of browsing. With this technique, the user's web browser is typically targeted for exploitation, but adversaries may also use compromised websites for non-exploitation behavior such as acquiring Application Access Token .
T1190	Exploit Public-	Adversaries may attempt to take advantage of a weakness in an Internet-facing computer or program

Two methods of **gaining initial access**:



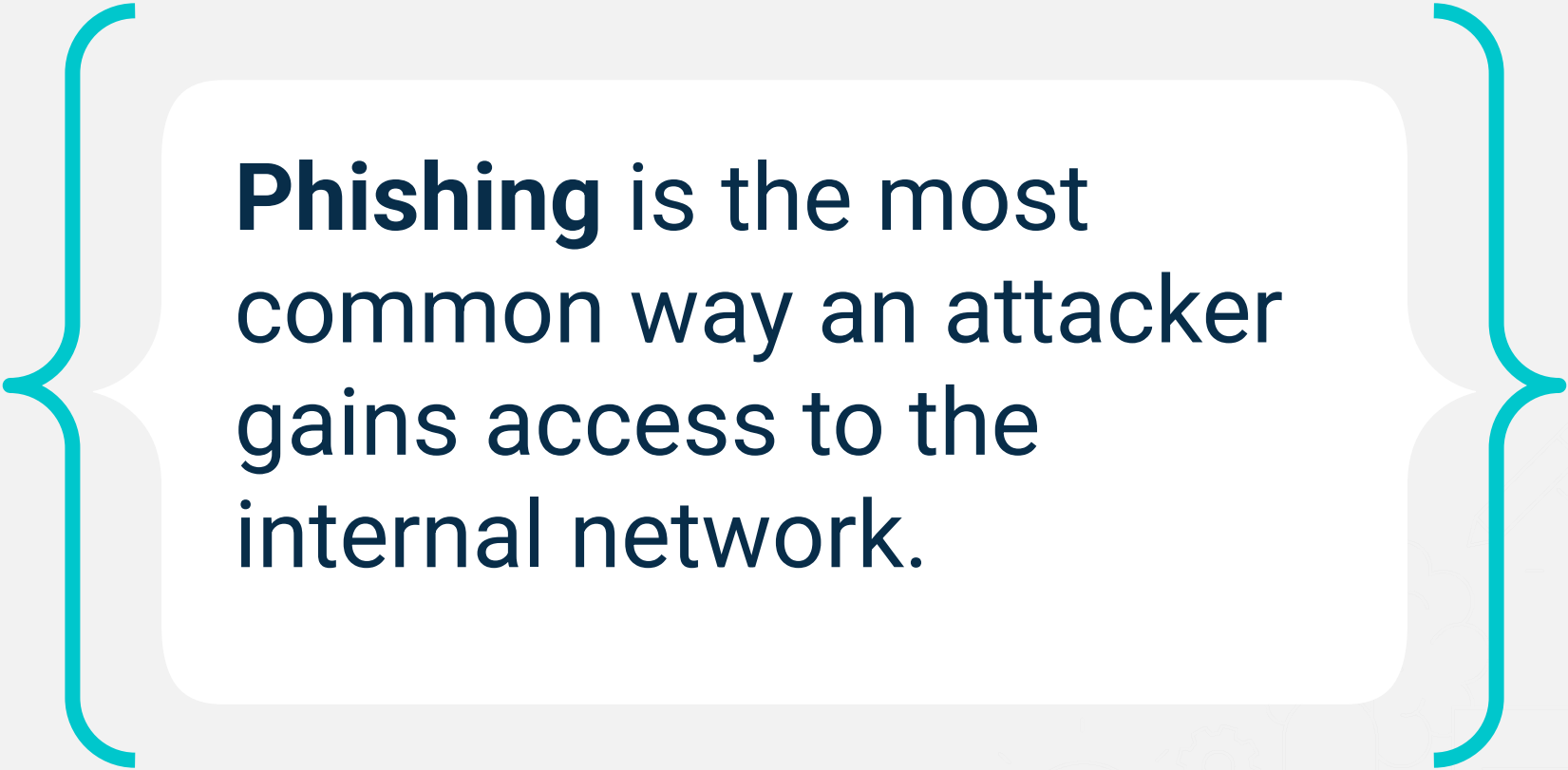
Phishing



**Remote Services with
Valid Accounts**



Phishing



Phishing is the most common way an attacker gains access to the internal network.





Phishing

The goal of phishing can vary



Sometimes,
a phishing email is sent only
to obtain credentials.

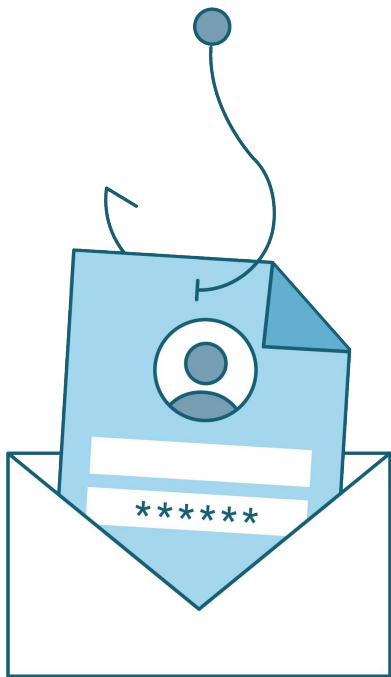


Other times,
the email contains a malicious link
that will install malware or a
backdoor on the victim's computer.



Phishing

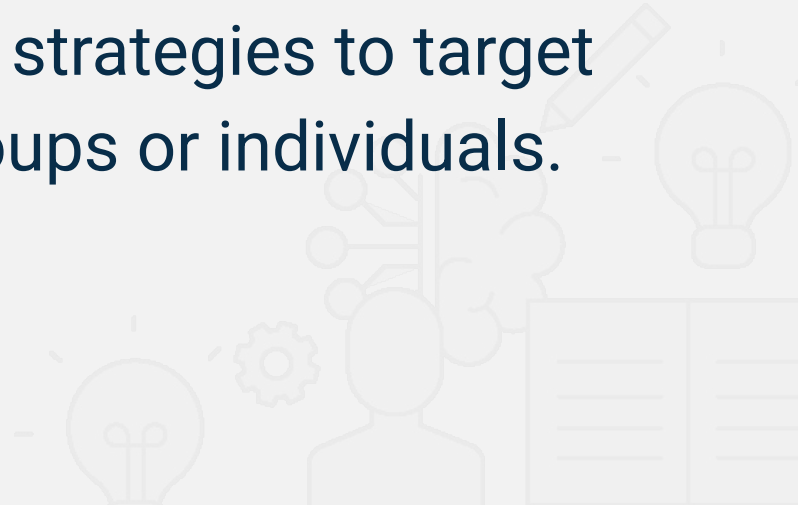
Phishing is more like a social engineering exercise than a technical abuse



- Phishing leverages human error
- No “patch” to prevent
- Best defense is **knowledge**



There are **different types** of phishing attacks. Attackers often adjust their strategies to target specific groups or individuals.





Spear Phishing

Attacker crafts a message to a specific individual or organization

Tactics:

Attackers use information gathered from social media, company websites, or other sources to personalize the email and make it more convincing.

Objective:

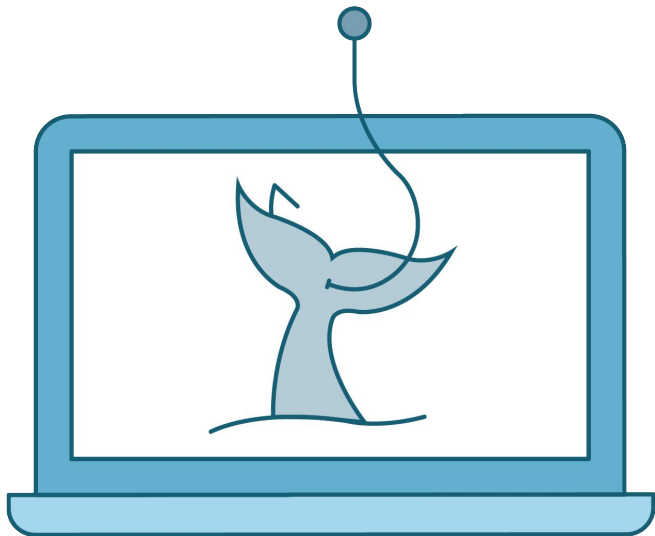
Have the targeted individual to divulge confidential information, click on a malicious link, or open an infected attachment.





Whaling

Aimed at senior executives or high-profile targets within an organization



Tactics:

The content of a whaling attack is often more sophisticated and may focus on executive-level issues.

Objective:

Given the high-profile nature of the targets, attackers aim to get large payments, sensitive corporate data, or other high-value returns.



Vishing

Attacks are carried out via phone calls

Tactics:

The attacker might pose as a bank representative, tech support agent, or any other official role to gain the victim's trust.

Objective:

The main goal is often to trick the victim into providing personal or financial information. This could include bank account numbers, passwords, or other sensitive data.





Has anyone ever encountered phishing attempts? What are some **telltale signs** that an email is a phishing attempt?

Phishing

Telltale signs of a phishing attempt:

Typos	Misspellings in the body of the email and the sender's email address are common red flags.
Typosquatting	A common tactic in which an attacker will register a fraudulent domain similar to a legitimate one. For example: Google.com VS. GoogIe.com (with a capital "i" instead of a lowercase "l")
Subdomains	Are often registered to mimic real domains. For example: payments.google.com VS. paymentsgoogle.bogusdomain.com Remember that domains can be spoofed if proper DNS protections are not in place.

Phishing

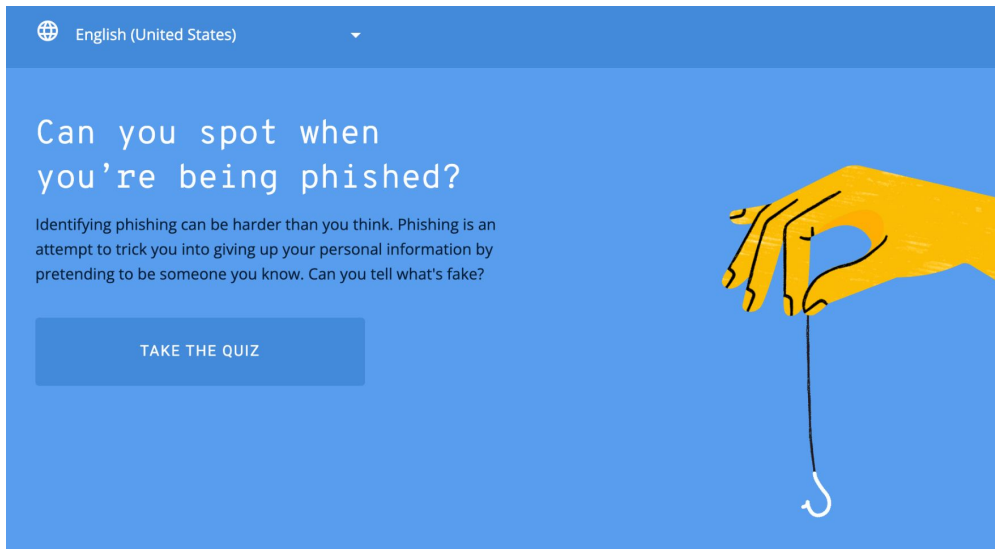
If an email asks you to do something that's not traditionally part of your job or that's possibly risky, it's always better to confirm the task in person or by another means of contact.

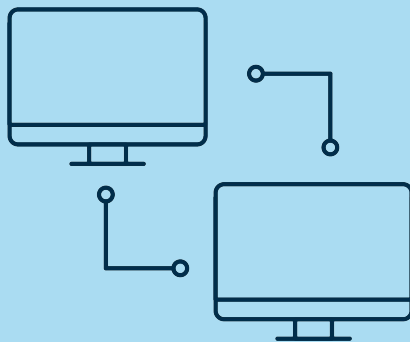


Phishing Quiz

It is not logistically feasible to perform a phishing campaign in class.
It is also illegal without written consent.

Instead, we'll practice
detecting the difference
between authentic and
phishing emails with an
online activity created
by Google.





Valid Accounts and External Remote Services

Valid Accounts and External Remote Services

Similar to a real attack, many different tactics may be attempted to gain access to a target during a penetration test.



Valid Accounts and External Remote Services

Another way of gaining initial access is by finding a VPN configuration file belonging to the target and logging into the VPN with stolen credentials. This method leverages two techniques from MITRE.

Valid Accounts

The attacker uses real user accounts to gain access.

Remote Services

The attacker uses remote services, such as VPN, to try and gain access from outside of the target's network.

Valid Accounts and External Remote Services

In one of our previous OSINT activities, we tried to find subdomains for `megacorpone.com`.

1

Suppose that during this process, we discovered an additional subdomain, `vpn.megacorpone.com`, that directly handles VPN connections and configurations for the MegaCorpOne domain.

2

We accessed this site and determined that it contains a login portal asking for a user id and password.

3

If a pen tester is able to supply correct credentials, they will be able to log in to the portal and download the VPN configuration file.

Valid Accounts and External Remote Services

While it's recommended that users and administrators create complex passwords, they often do not.

Due to this vulnerability, it is not uncommon for pen testers to attempt and succeed at logging into accounts using weak, commonly used passwords.

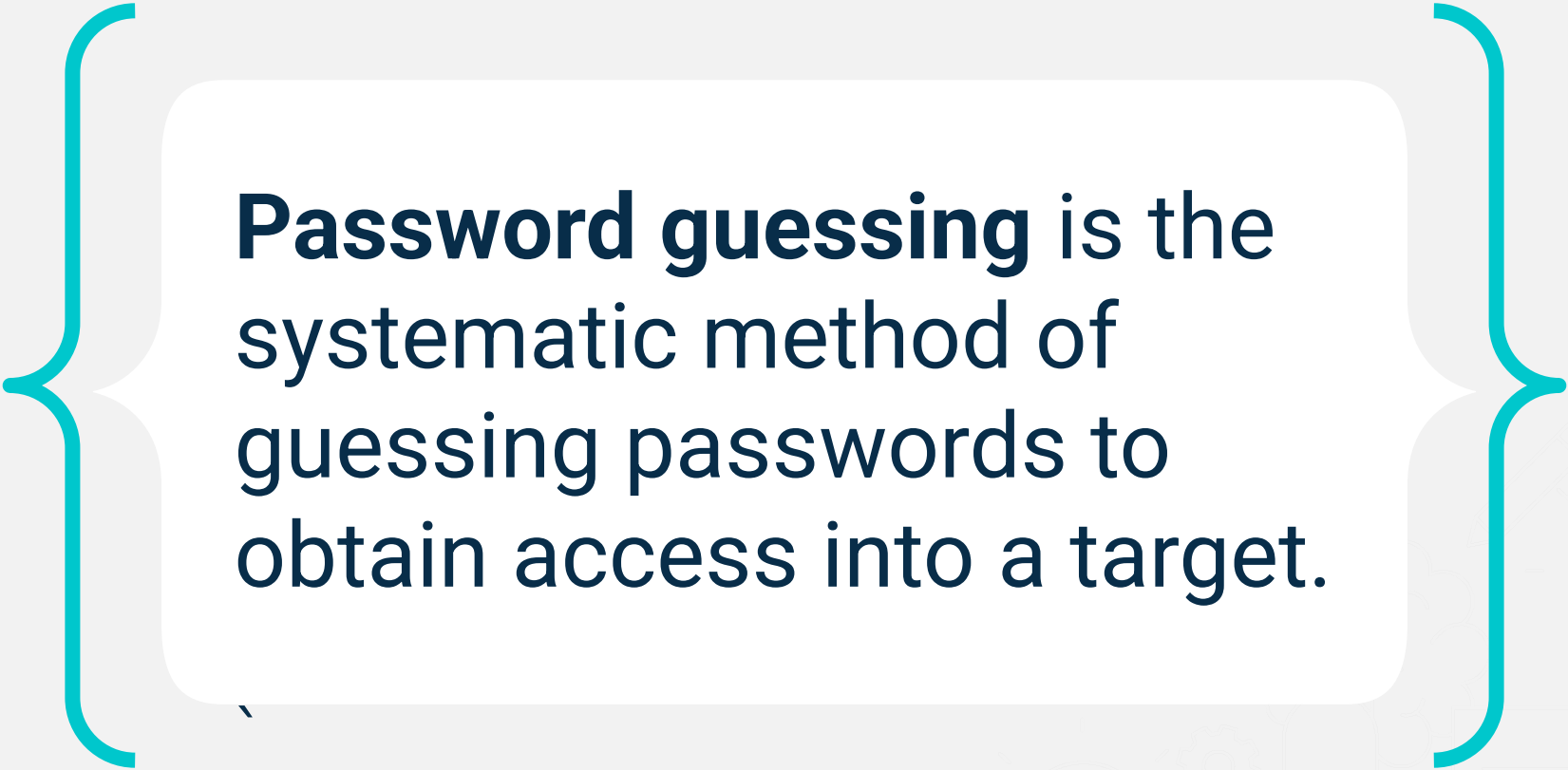
The method they would apply is called **password guessing**..

Password

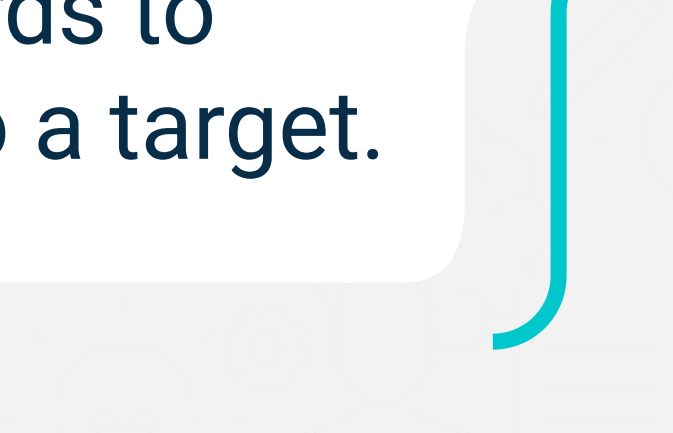
 Weak

Password

 Strong

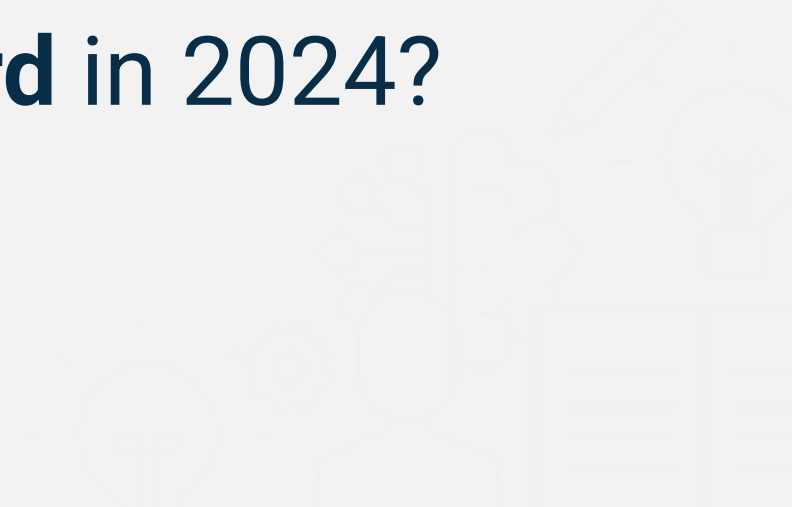


Password guessing is the systematic method of guessing passwords to obtain access into a target.





Can you guess the
most commonly used
password in 2024?



The most common password was

123456

You can visit [this website](#) for a list of the other most commonly used passwords.

Password Guessing

Many common passwords include the following:

Variants of “password”	(P@ssw0rd, pa\$\$w0rd, etc.)
Season/Year	(Winter2020, Summer2021)
The user’s own username or variants of it	(bob, b0b, a\$h13y, etc.)

This information can help pen testers succeed in correctly guessing a user’s password.

Password guessing has also been used in high-profile breaches such as the **SolarWinds hack**.

During this attack, threat actors used password guessing to breach their targets.





Let's recap

Summary

We've covered the following concepts:

Initial Access

MITRE tactic covering methods for gaining access into a target's system.

Phishing

Most commonly used initial access method.

Typosquatting is a tactic where an attacker will register a fraudulent domain similar to a legitimate one.

Remote services through valid accounts

Initial access method where the attacker uses real user accounts to gain access.

Used in conjunction with **password guessing**.



Questions?





Activity:

Accessing Remote Services through Valid Accounts

In this activity, you will continue to play the role of a pen tester conducting an engagement on MegaCorpOne.

- One of MegaCorpOne's main concerns is that its password security is very poor and allows weak passwords.
- You are tasked with using the usernames that you found during the Day 1 OSINT phase to try to log in to `vpn.megacorpone.com` via **password guessing**.
- Once logged in, you will run a shell script that will log you in to the internal network.

Suggested Time:
15 Minutes

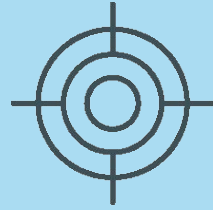


Time's up!
Let's review



Questions?

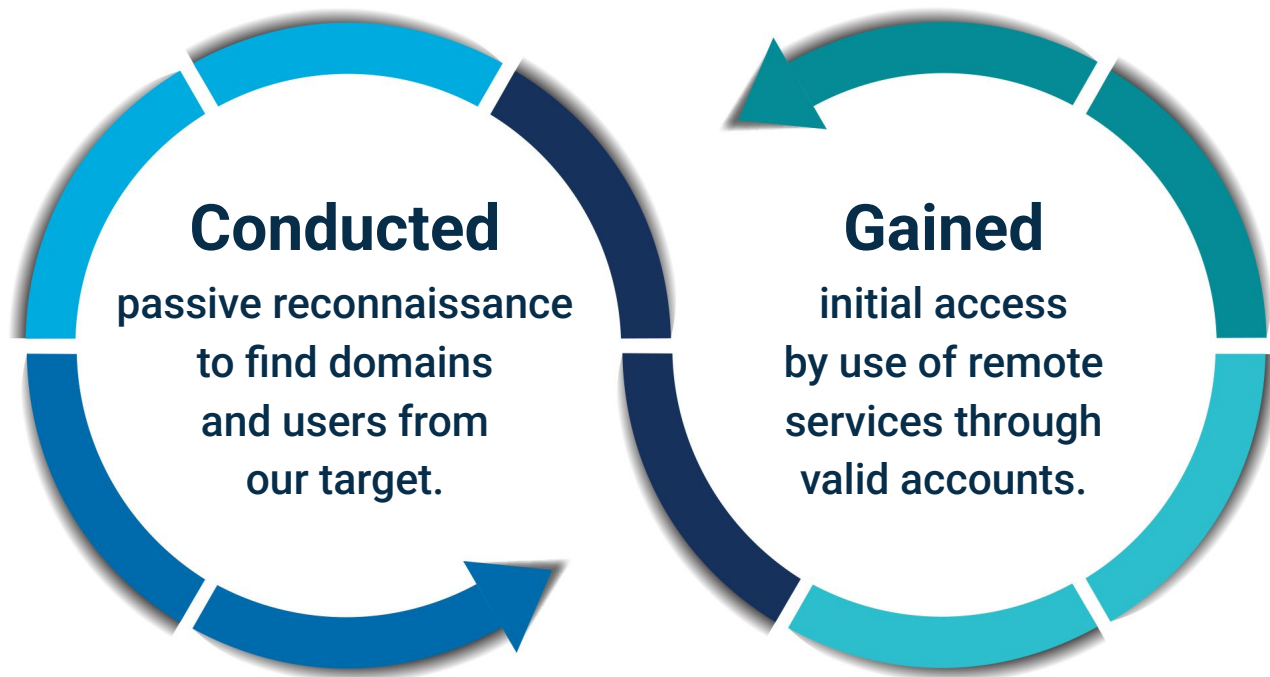




Scanning and Internal **Reconnaissance**

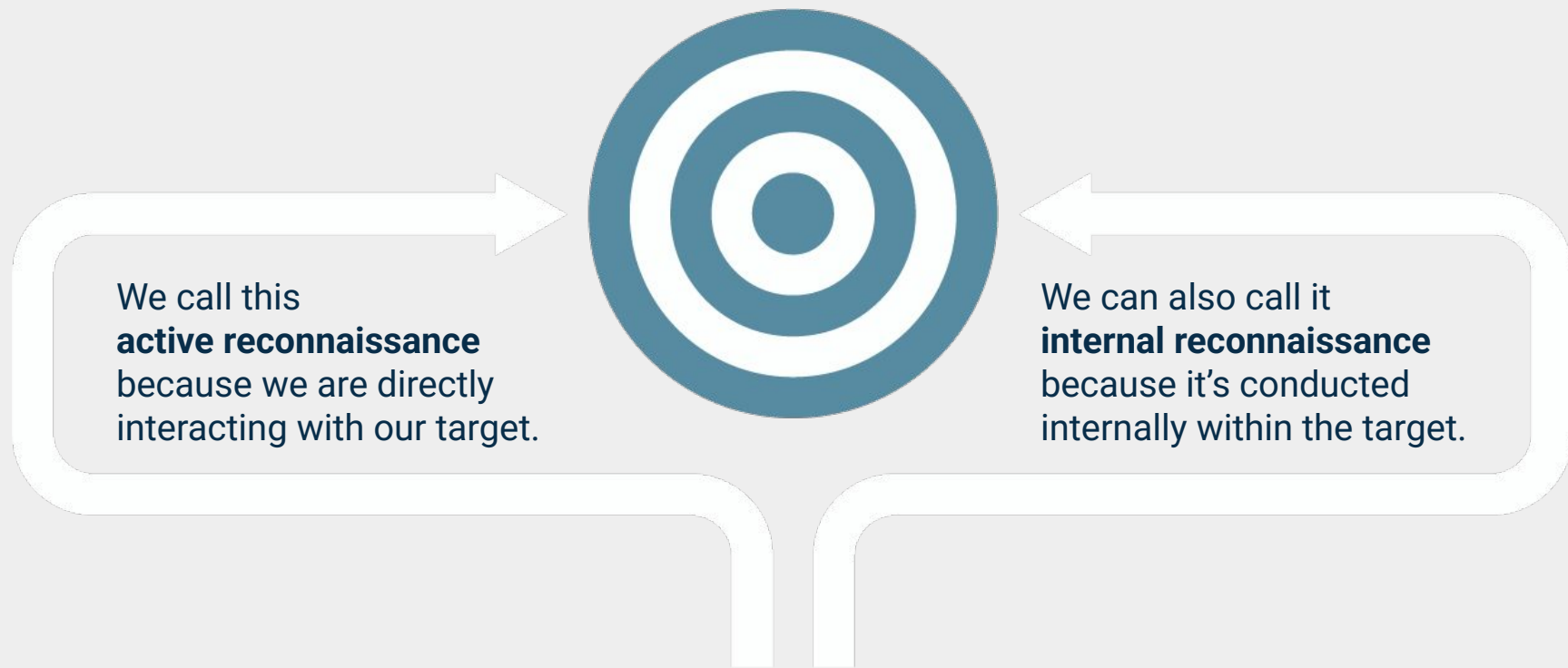
Scanning and Internal Reconnaissance

So far we have:



Internal Reconnaissance

Now, we'll perform reconnaissance inside the internal network, known as **active reconnaissance**, to reveal which devices are on the network and what potential new targets await.



Internal Reconnaissance

Inside the network, we can gather the following:



**Host
Enumeration**



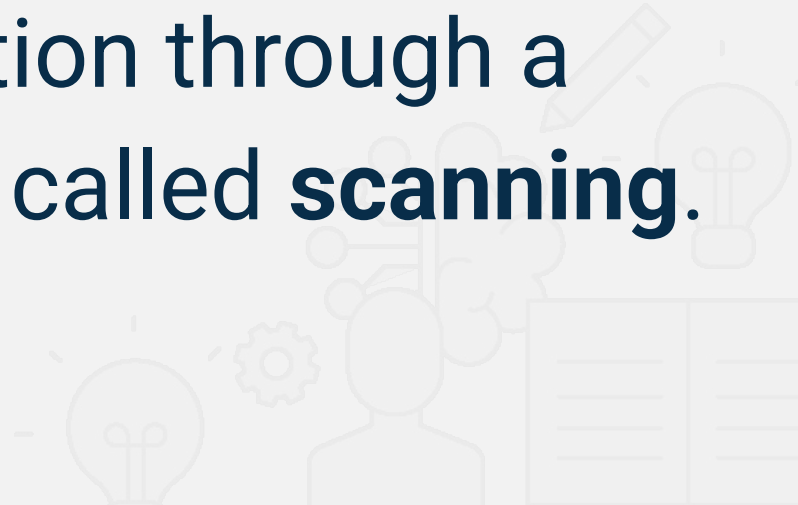
**Process
Enumeration**



**Information on
Users**

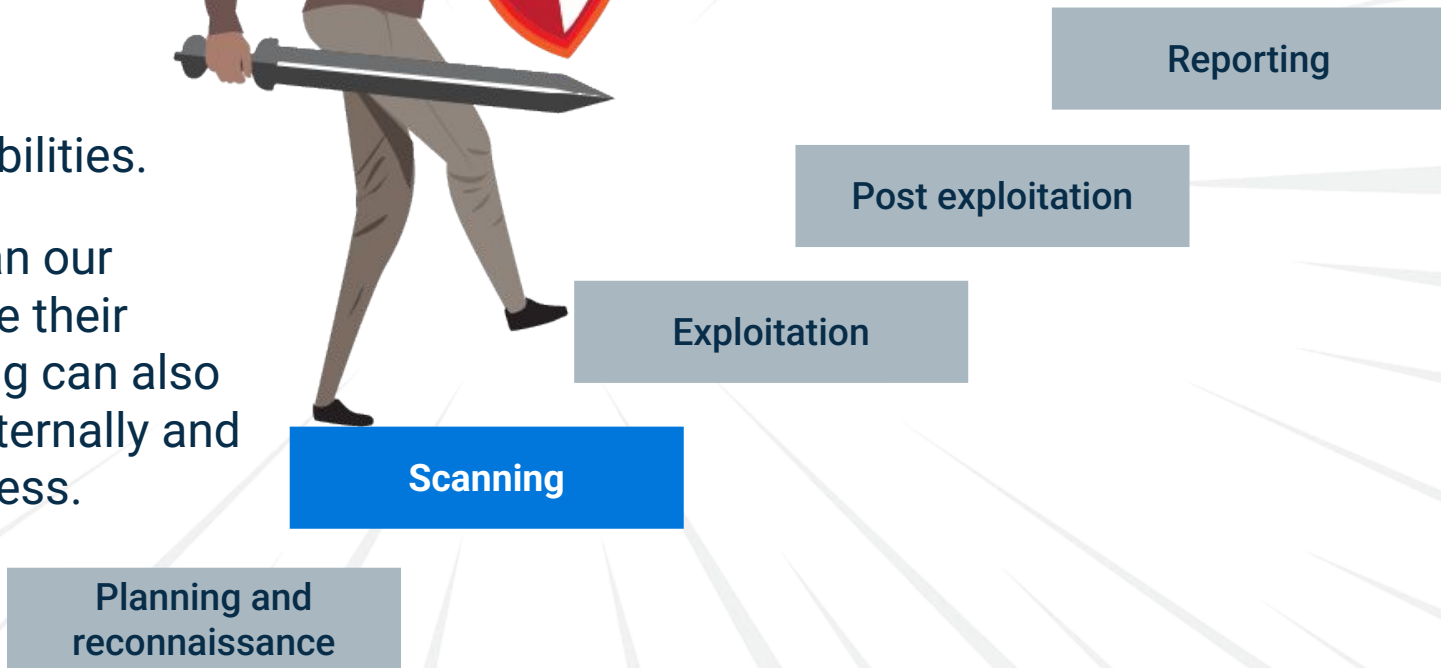


We can more aggressively gather information through a process called **scanning**.



Scanning is the second phase of pen testing. Scanning uses tools to gather information such as network information and potential vulnerabilities.

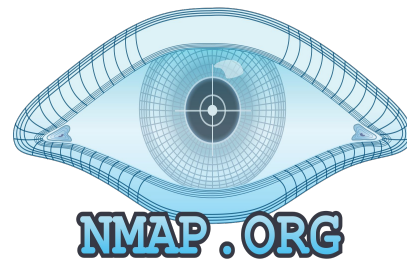
While we will scan our target from inside their network, scanning can also be conducted externally and before initial access.





Phase 2: Scanning

The following tools are often used to conduct scanning:



We will explore the scanning phase by revisiting the tool Nmap.

Nmap

Nmap is a popular scanning tool that we covered in the earlier Networking modules

Nmap excels at identifying which ports are open, the services behind those ports, and their versions.

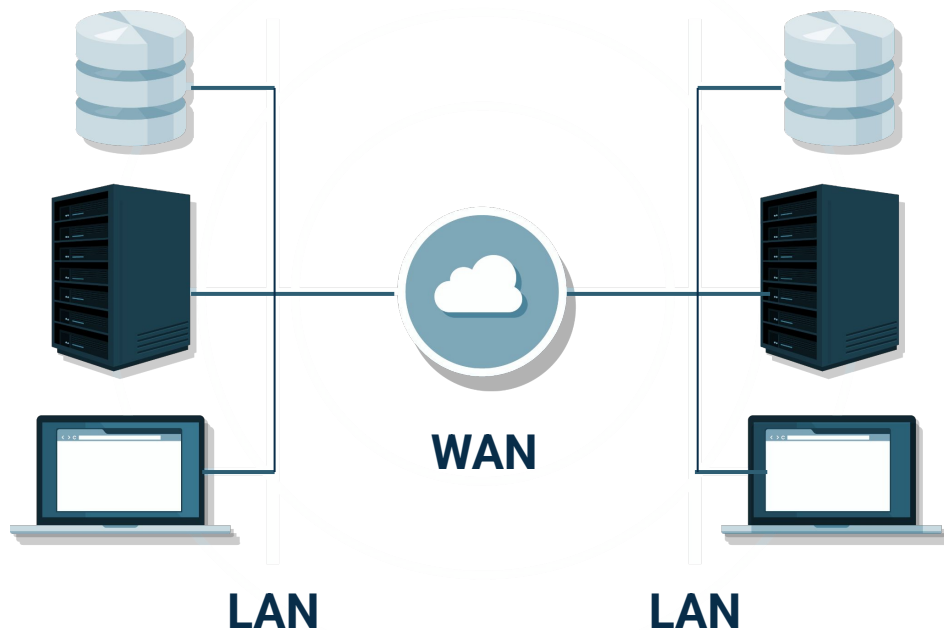
Nmap has additional functionality that we will explore in a later activity.



Scanning and Internal Reconnaissance

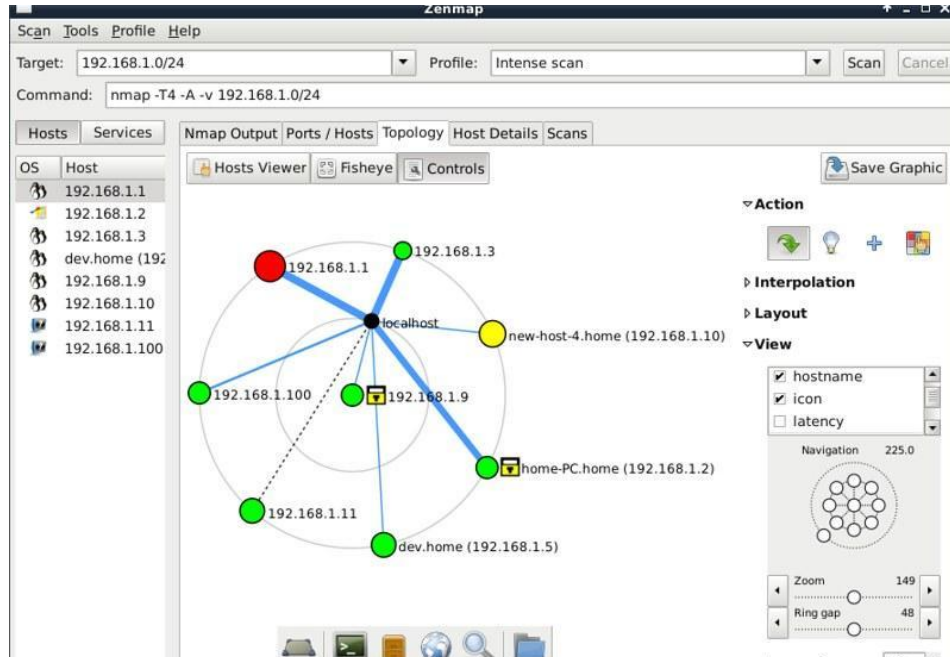
It's important to identify the purpose of the machines on the network. For example:

- A properly set-up network has several virtual LANs (VLANs) with different purposes (e.g., servers on one, workstations on another).
- Identifying **high-value targets** (secret-storing servers, domain controllers, etc.) is important, as they're often an end-game target.
- Servers often have less security software on them than user workstations, because businesses focus on preventing attackers from gaining initial access.



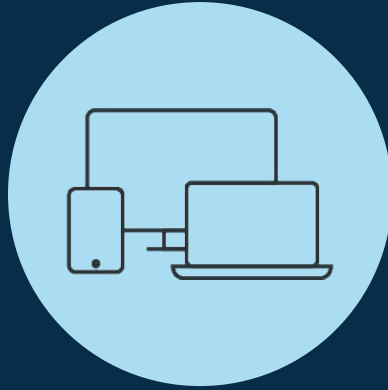


Demonstration Scenario



In the following demonstration using Zenmap and Nmap, we will:

- Use Nmap to perform a port scan on an internal network.
- Introduce a GUI alternative to Nmap called Zenmap.
- Use Zenmap to perform a port scan and highlight the additional features offered.



Instructor Demonstration

Nmap and Zenmap

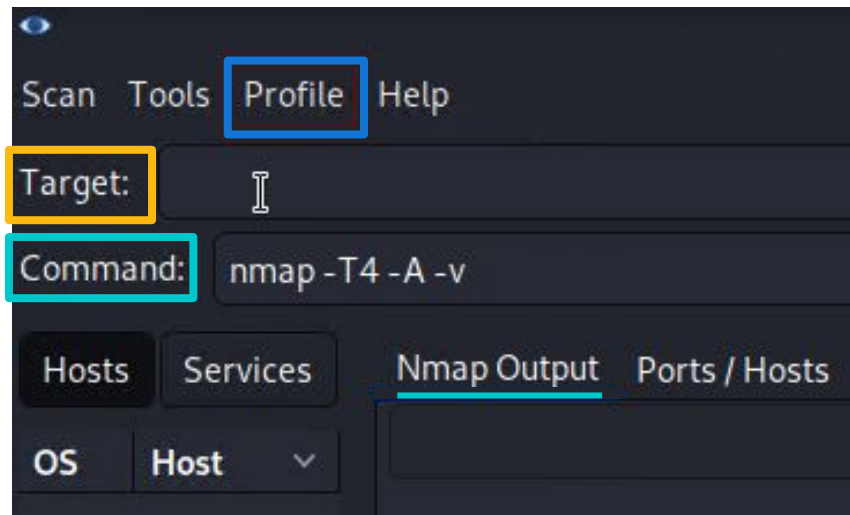
Nmap and Zenmap Demonstration

The Zenmap interface includes:

Target: Where we input a hostname or IP address. CIDR notation is also accepted.

Profile: A drop-down menu that contains several pre-built scans. These can be changed and edited. You can also create custom profiles.

Command: Shows the Nmap command being run. This will update automatically if we change or edit a profile.



Summary

-  **Scanning** is the second phase of a pen testing engagement
-  Scanning is often conducted **externally**, before initial access.
-  Nessus, Hping, and Nmap are tools that we often use to conduct scanning.
-  Zenmap is the GUI version of Nmap.
-  NSEs are scripts that are used to test if a service is vulnerable to an exploit.



Activity:

Zenmap

In this activity, you will continue to play the role of a pen tester conducting an engagement on MegaCorpOne.

- Now that you are on the internal network, you'll get "the lay of the land" by performing scanning.
- You will use Zenmap and NSE scripts to build out a profile in order to perform a scan on a certain machine on the network.

Suggested Time:
10 Minutes



Time's up!
Let's review



Questions?





Vulnerability Scanning



Vulnerability Scanning

NSE allows users to modify and create custom Nmap scripts



While NSE has its advantages, it also has **disadvantages** when compared to other vulnerability scanners:

- Not fully comprehensive
- Cannot perform a large number of scans simultaneously
- Most efficient when performing single host scans
- Most useful when doing basic information gathering or enumeration activities



Vulnerability scanners
can help make up for
many of the limitations
of NSE.



Vulnerability Scanning

Pen Testing vs. Vulnerability Scanning

Vulnerability Scanning

Identifies systems that have known vulnerabilities.

- Scans use a database of known vulnerabilities.
- Vulnerabilities are rated based on the severity level.
- Vulnerabilities are given a Common Vulnerability Scoring System (CVSS) score.

Penetration Testing

Attempts to identify weaknesses that can be exploited, such as:

- Specific system configurations
- Organizational processes and practices



Vulnerability Scanning

Pen Testing vs. Vulnerability Scanning

Unfortunately, companies often do not understand the differences between the two and sometimes think a vulnerability scan **passes as a penetration test** when they're actually very different.

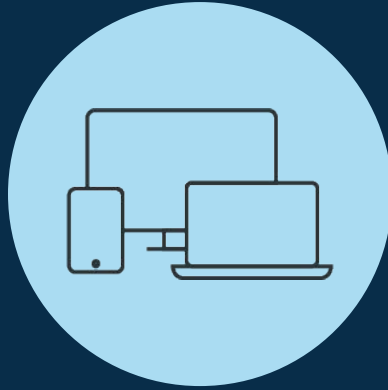


Vulnerability Scanning and Nessus

A vulnerability scanner, such as Nessus, is an application that identifies vulnerabilities and creates inventory of all interconnected systems.

These include the following:

- | | | | |
|----|------------------|----|------------|
| 01 | Servers | 05 | Containers |
| 02 | Desktops | 06 | Firewalls |
| 03 | Laptops | 07 | Switches |
| 04 | Virtual machines | 08 | Printers |



Instructor Demonstration

Nessus

Summary

- **Nessus** is a free, open-source network vulnerability scanner.
- Nessus discovers vulnerabilities that malicious hackers can exploit to breach a network, and it generates reports.
- Vulnerability scans search for known vulnerabilities in a system and report potential exposures.
- Penetration tests weaknesses in network infrastructure to determine the level at which a malicious actor can gain unauthorized access.

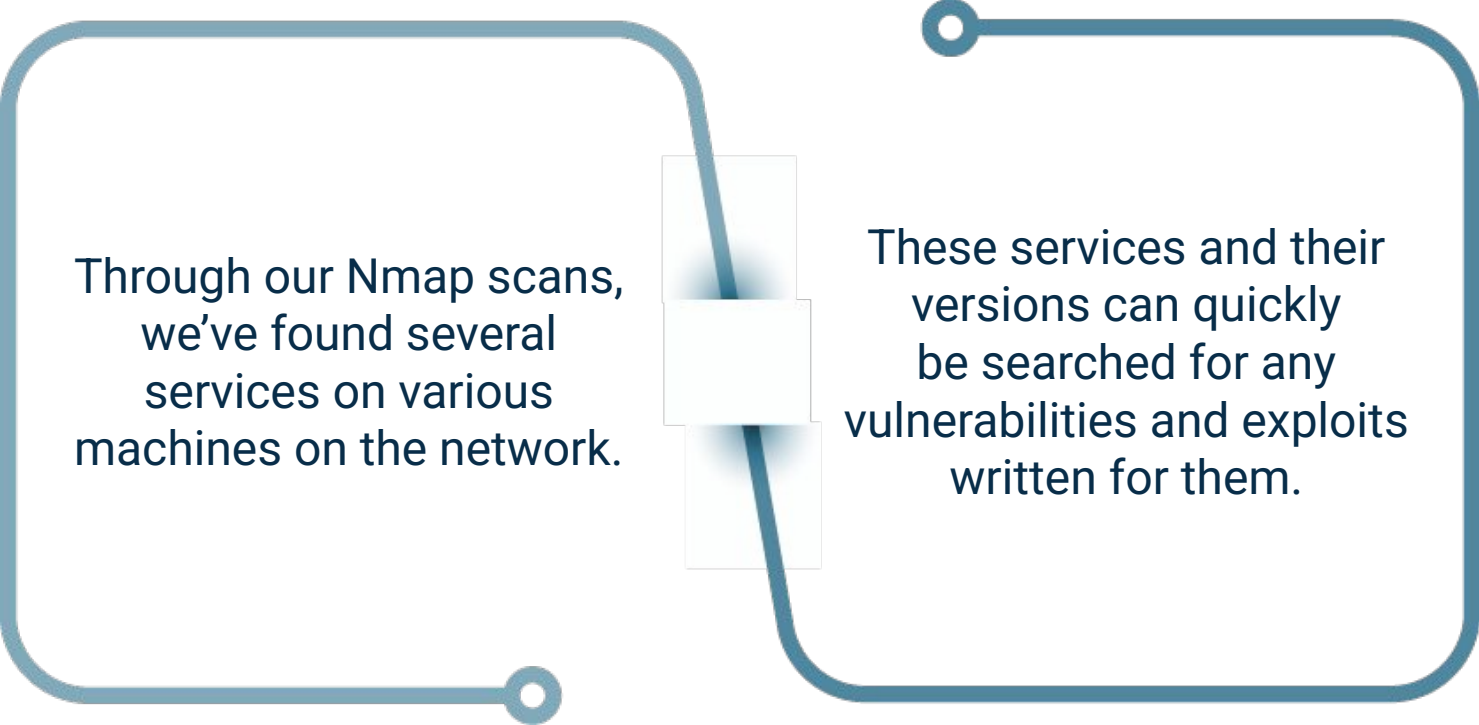


Break
15 mins



SearchSploit

SearchSploit



Through our Nmap scans,
we've found several
services on various
machines on the network.

These services and their
versions can quickly
be searched for any
vulnerabilities and exploits
written for them.

SearchSploit

Now, we will use a tool called **SearchSploit** to locally store a library of exploit information, as **Exploit Database (Exploit DB)** is an open-source exploits database that contains publicly disclosed exploits, cataloged according to their **Common Vulnerability and Exposure (CVE)** identifiers.



```
File  Actions  Edit  View  Help
```

```
kali@kali:~$ searchsploit
```

```
Usage: searchsploit [options] term1 [term2] ... [termN]
```

```
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```

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Examples
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```

```
searchsploit afd windows local
```

```
searchsploit -t oracle windows
```

```
searchsploit -p 39446
```

```
searchsploit linux kernel 3.2 --exclude="(PoC)|/dos/"
```

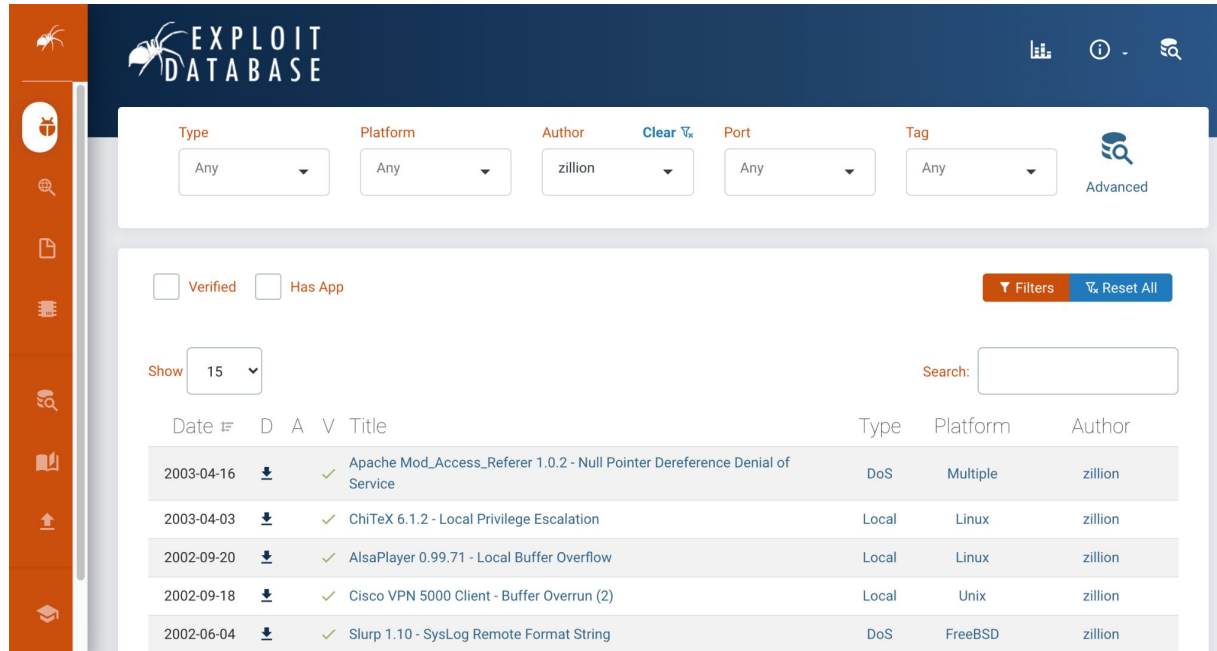
```
searchsploit -s Apache Struts 2.0.0
```

```
searchsploit linux reverse password
```

```
searchsploit -j 55555 | json_pp
```

SearchSploit

Exploit-DB's catalog of exploits is constantly updated as software developers patch their systems.



The screenshot displays the Exploit-DB SearchSploit interface. The top navigation bar is dark blue with the 'EXPLOIT DATABASE' logo and icons for a bar chart, information, and search. Below this is a filter section with dropdown menus for Type (Any), Platform (Any), Author (zillion), Port (Any), and Tag (Any), along with a 'Clear' button and an 'Advanced' search icon. Further down, there are checkboxes for 'Verified' and 'Has App', a 'Filters' button, and a 'Reset All' button. A 'Show' dropdown is set to 15, and a search input field is present. The main content area is a table of exploits with columns for Date, D (download), A (add), V (verified), Title, Type, Platform, and Author.

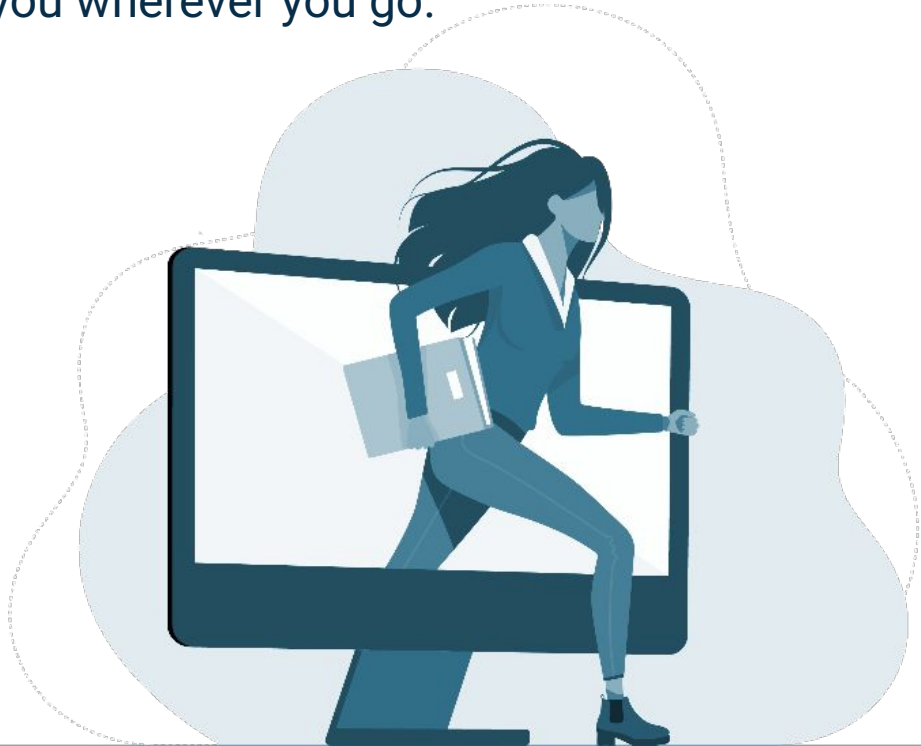
Date	D	A	V	Title	Type	Platform	Author
2003-04-16	↓		✓	Apache Mod_Access_REFERER 1.0.2 - Null Pointer Dereference Denial of Service	DoS	Multiple	zillion
2003-04-03	↓		✓	ChiTeX 6.1.2 - Local Privilege Escalation	Local	Linux	zillion
2002-09-20	↓		✓	AlsaPlayer 0.99.71 - Local Buffer Overflow	Local	Linux	zillion
2002-09-18	↓		✓	Cisco VPN 5000 Client - Buffer Overrun (2)	Local	Unix	zillion
2002-06-04	↓		✓	Slurp 1.10 - SysLog Remote Format String	DoS	FreeBSD	zillion

SearchSploit Overview

SearchSploit is a command-line utility for Exploit-DB that allows you to take an offline copy of the Exploit-DB with you wherever you go.

Security professionals can perform detailed offline searches of hundreds of exploit scripts through their local copy of the repository.

This capability is useful if you are working on a security assessment with an air-gapped, segregated network that lacks internet connectivity.



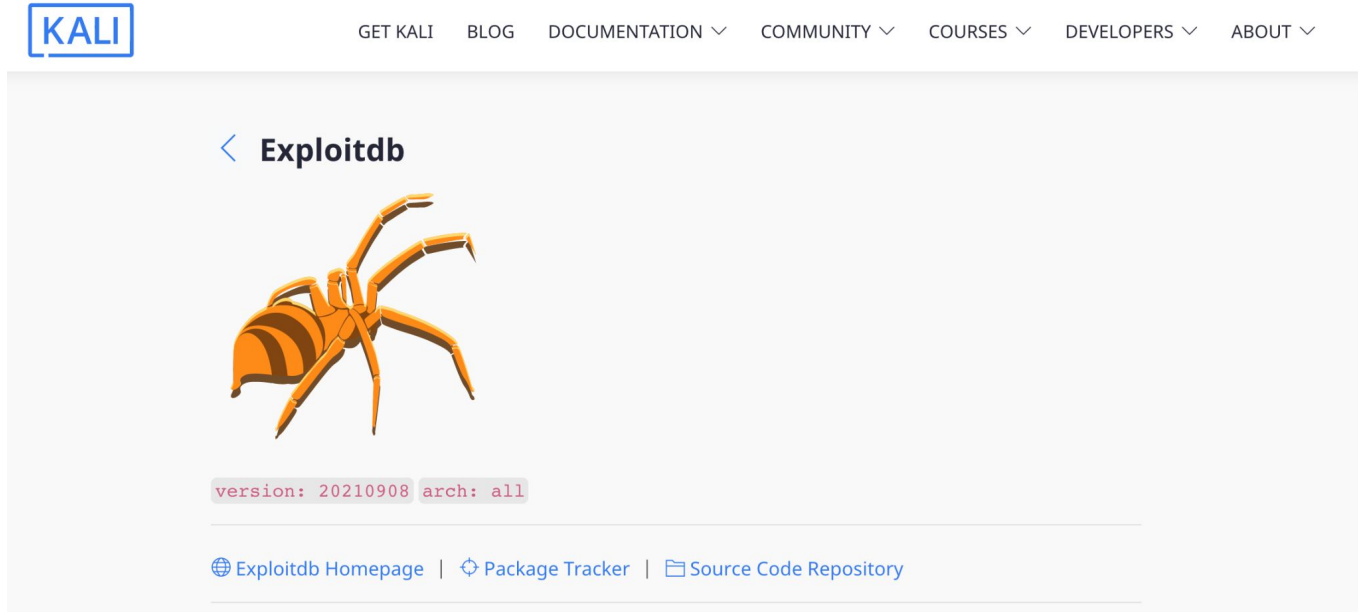
SearchSploit Overview

SearchSploit, as indicated by its name, will search for all exploits and shell code contained within the Exploit-DB repository.



SearchSploit Overview

SearchSploit comes preinstalled on Kali Linux, but should be updated on a weekly basis and prior to each use.

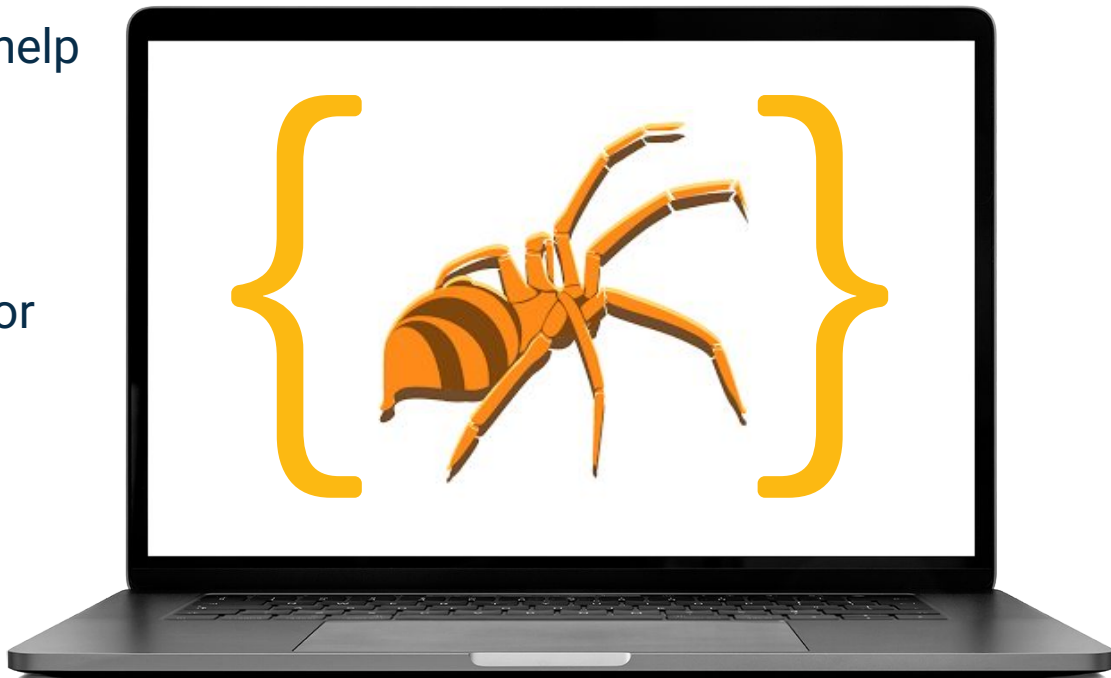


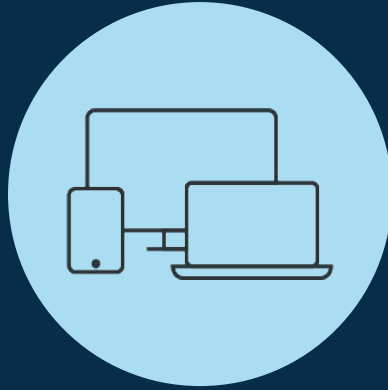


Demonstration Scenario

In this demonstration, we will:

- Run through some basic help commands
- Discuss the various file formats
- Cover command syntax for SearchSploit





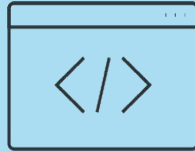
Instructor Demonstration

SearchSploit



Questions?





Exploitation with Shells

Shells

A common goal of exploitation is to obtain access to the remote machine.

1

Exploited access to the machine is typically granted in the form of terminal access, which is referred to as a **shell**.

2

A shell is a **terminal**. Opening a terminal on your Kali machine is opening a “shell” on your own machine.

3

A **remote shell** is opened on a remote computer on the network.

Shells

We will cover two types of shells today:

Bind shell

The remote host opens a port for the current host to connect to.

The current, local host then connects to that remote host's port.

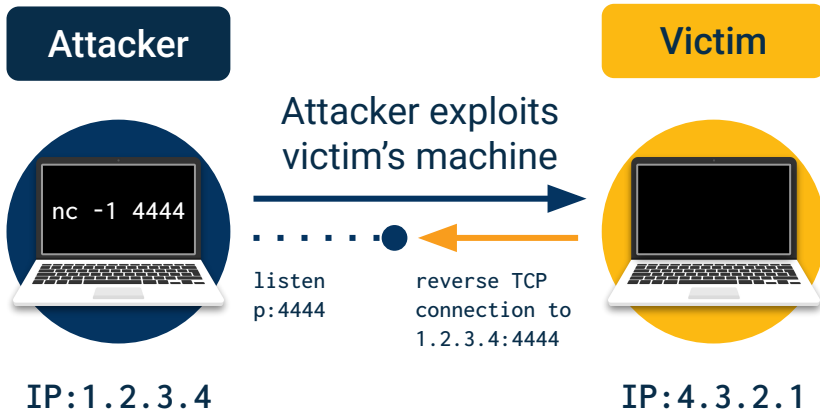
Reverse shell

The remote host connects back to a port on the local host.

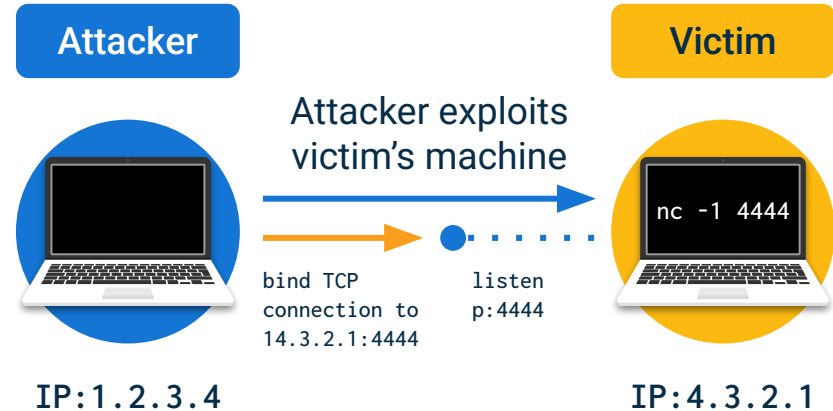
Shells

The advantage of a reverse shell over a bind shell is that **egress** (exit) ports are more commonly open than **ingress** (enter) ports. This means that a remote host will have fewer firewall issues connecting back to a port on the local host. This also circumvents NATing.

Reverse shell



Bind shell



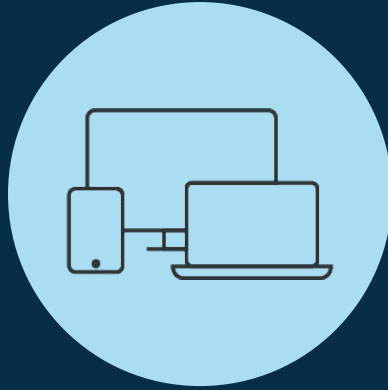


Demonstration Scenario



Next, we'll use a tool called **Netcat** to demonstrate bind and reverse shells.

Netcat is commonly referred to as the “**Swiss army knife**” of networking tools, because it can assist with many security and admin activities.



Instructor Demonstration

Shells

Summary

- **Exploit-DB.com** is a popular online database that contains publicly disclosed exploits, cataloged according to their CVE identifiers.
- **SearchSploit** is a command-line utility for Exploit-DB that allows you to take an offline copy of the Exploit database with you wherever you go.
- Exploited access to a machine is typically in the form of a shell.
- Two types of shells are bind and reverse.



Activity:

Exploitation

In this activity, you will continue to play the role of a pen tester conducting an engagement on MegaCorpOne.

- Now that you have identified a potential exploitable service on a host, you will use SearchSploit to find a script that can exploit it.
- You will then use a SearchSploit exploit to determine whether you can gain shell access on the host.

Suggested Time:
20 Minutes



Time's up!
Let's review



Questions?





Next class

Exploring Exploitation



Thank you